

Bhagya Majithiya

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Education

Dhirubhai Ambani University
B.Tech in Information and Communication Technology – CPI: 7.09

Gandhinagar, Gujarat
Aug 2023 – May 2027

Experience

Smart Energy Learning Centre, Dhirubhai Ambani University
Research Intern

Gandhinagar, Gujarat
May 2026 – Jul 2026

- Engineered an IoT sensor-fusion prototype on **STM32/C2000** integrating **IMU, magnetometer, and GPS** for real-time position and heading estimation.
- Designed and validated an **Extended Kalman Filter (EKF)** in **MATLAB/Simulink** to fuse accelerometer, gyroscope, and GPS data for drift correction and navigation in GPS-denied environments.

Projects

LeetFlowAI – AI-Powered DSA Coach | Next.js, TypeScript, Gemini, Prisma, PostgreSQL, Clerk, Recharts

- Built a full-stack AI DSA coaching platform with **live LeetCode profile synchronization via GraphQL**, contest analytics, topic-wise insights, and personalized DSA roadmaps.
- Integrated **Google Gemini** to generate optimized solutions with **Big-O time and space analysis** across **9+ programming languages**; used TanStack Query for efficient server-state caching.
- Architected the backend using **Prisma ORM with PostgreSQL**, Clerk authentication, Recharts dashboards, and CI/CD deployment on **Vercel**.

PlanIt – AI Task Scheduler | Next.js, TypeScript, MongoDB, Gemini AI, NextAuth, Node.js

- Developed a full-stack task management platform with **CRUD operations, deadline tracking, priority management, reminders, CSV export**, and progress visualization.
- Integrated a **Gemini-powered chatbot** with persistent multi-thread conversations for natural-language task input, priority suggestions, and productivity summaries.

Mini-RAG – Retrieval-Augmented Generation System | Python, Google Gemini, FAISS, Sentence Transformers, PyPDF

- Built a modular **RAG pipeline** for document question answering with separate ingestion and retrieval stages, enabling reusable vector indexes.
- Implemented document parsing, chunking, **Sentence Transformer embeddings**, and **FAISS similarity search**, with Gemini generating context-grounded responses.

DualTrack – EKF-SLAM vs Dead Reckoning | Kotlin, Android SDK, EKF, IMU, OpenCV, Computer Vision

- Built an Android application comparing **EKF-SLAM and Dead Reckoning** localization using smartphone accelerometer, gyroscope, and camera sensors without external infrastructure.
- Implemented the EKF predict-update cycle with **Jacobian computation, data association, and covariance estimation** to track position and landmarks.
- Fused visual localization with Dead Reckoning using a complementary filter and visualized real-time trajectories, landmarks, and localization error metrics on-device.

Technical Skills

Languages: C, C++, Python, TypeScript, JavaScript, SQL, MATLAB, HTML/CSS, Kotlin

Frameworks & Libraries: Next.js, React, Node.js, Express.js, LangChain, HuggingFace Transformers, Prisma, Mongoose, Sentence Transformers, OpenCV

Tools & Platforms: Git, GitHub, Linux Shell, Postman, VS Code, MATLAB Simulink, Google Colab, Vercel, Docker

Cloud & Databases: MongoDB, PostgreSQL, MySQL, Supabase, Google Cloud Platform, Firebase

Concepts: Data Structures and Algorithms, Object-Oriented Programming, System Design, RAG, AI-ML, EKF/SLAM, IoT, Cloud Computing, Computer Networks, EDA

Achievements & Competitive Programming

- JEE 2023 – 98.22 Percentile:** Ranked in the **top 2%** among candidates nationally.
- LeetCode:** **350+ problems** solved (q0EfC7Dwsj) | Ranked **4,255 / 46,224 (Top 9.2%)** in Weekly Contest 516.
- CodeChef:** Competitive Rating **1552** | DSA Rating **1869** (bhagya_1100) | Ranked **229 / 6,875 (Top 3.3%)** in the DSA Monday 017 AK Contest.